

6. Neural Networks III

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D MATH

In the previous lectures we saw...

- The Universal Approximation Property (UAP) for neural networks
- Complexity bounds for shallow and deep networks

But...

- UAP proofs and complexity bounds are only existence results
- In practice we train networks using SGD
- We would like to:
 - Study the dynamics of the neural network during training
 - Understand the generalization properties of the solution (not just bias, also variance!)

In today's class...

6.8 The generalization puzzle

6.9 The Neural Tangent Kernel

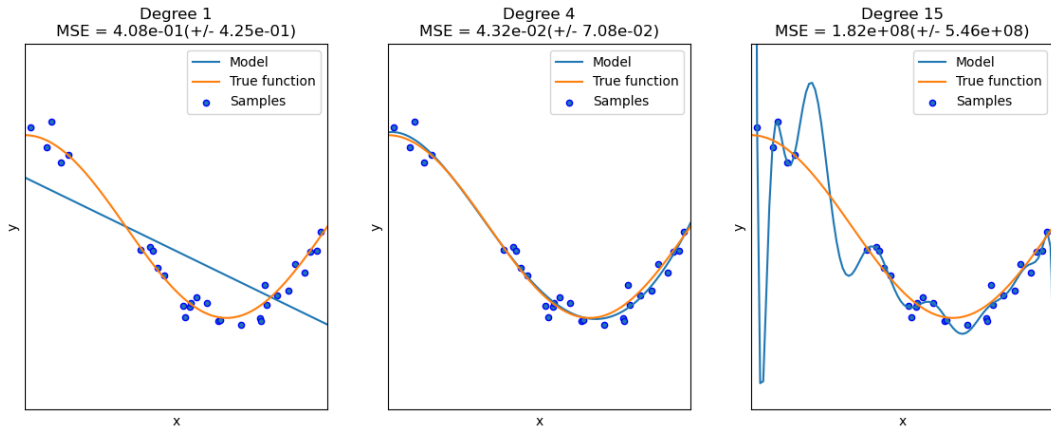
- Definition
- Convergence at initialization
- Convergence during training

6.10 Convergence to the minimum-norm interpolator

6.11 Generalization in the overparametrized regime

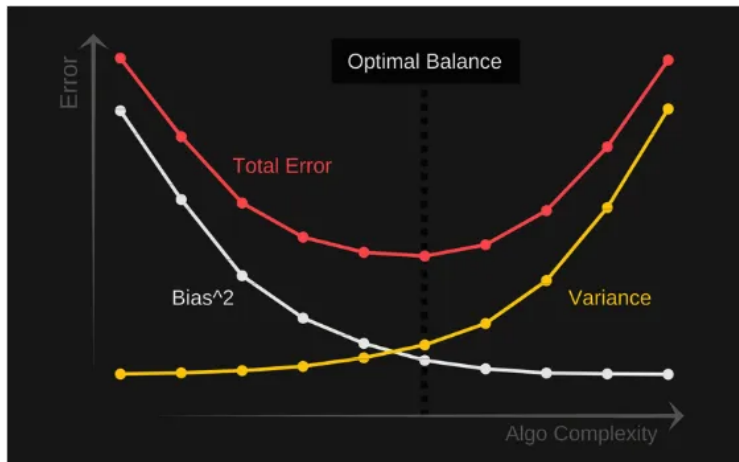
6.8 The generalization puzzle

Recall the classical bias-variance trade-off for polynomial regression (with test MSE):



- Low model complexity causes underfitting and poor generalization (bias is too high).
- High model complexity causes overfitting and poor generalization (variance is too high).
- There is a trade-off between bias and variance.

6.8 The generalization puzzle



6.8 The generalization puzzle

In [Zhang et al., 2017] they trained a (convolutional) neural network with 1.6 million parameters on a dataset of 50,000 images (very large model, overparametrized regime).

Task: classify each image into one of ten different classes.



They showed that:

- the model achieved zero training error, but had state-of-the-art generalization performance
⇒ overfitting does not always cause poor generalization!
- the model achieved zero training error also on training data with randomized labels
⇒ the model is complex enough to interpolate any dataset!
- increasing the size of the model could lead to even lower generalization error
⇒ variance actually decreases with model complexity!

6.8 The generalization puzzle

- The ability of overparametrized neural networks to generalize well - contrary to what we know from the classical bias-variance trade-off - is often referred to as the ***generalization puzzle***.
- Their results have been replicated empirically in many settings (see [Neyshabur et al., 2014], [Neyshabur et al., 2017], [Belkin et al., 2019])
- Various explanations have been proposed:
 - Alternative measures of complexity for neural networks (local Rademacher complexity [Bartlett et al., 2005], norm-based complexity bounds [Bartlett, 1998])
 - Alternative approaches to generalization (e.g. algorithmic stability [Bousquet and Elisseeff, 2002], predictivity [Poggio et al., 2004]), compression bounds [Arora et al., 2018], algorithmic robustness [Xu and Mannor, 2012])
 - Optimization and implicit regularization (see [Dziugaite and Roy, 2017] [Neyshabur et al., 2017])
- Today we focus on a connection between neural networks and RKHS, based on the concept of ***Neural Tangent Kernel*** [Jacot et al., 2018], which helps us understand these empirical findings.

6.9 Neural Tangent Kernel: definition

Notation and calculus recap:

- If $f : \mathbb{R}^m \rightarrow \mathbb{R}^n$ is a differentiable function, its Jacobian is defined as:

$$Df(\theta) := \begin{pmatrix} \nabla f_1(\theta)^T \\ \vdots \\ \nabla f_n(\theta)^T \end{pmatrix} = \begin{pmatrix} \frac{\partial}{\partial \theta_1} f_1(\theta) & \dots & \frac{\partial}{\partial \theta_m} f_1(\theta) \\ \vdots & \ddots & \vdots \\ \frac{\partial}{\partial \theta_1} f_n(\theta) & \dots & \frac{\partial}{\partial \theta_m} f_n(\theta) \end{pmatrix} \in \mathbb{R}^{n \times m}$$

- For functions of two variables $(\theta, \phi) \mapsto f(\theta, \phi)$, we denote by $D_\theta f(\theta, \phi)$ the Jacobian of f with respect to the variable θ evaluated at (θ, ϕ) .
- **(Chain rule)** If $f : \mathbb{R}^m \rightarrow \mathbb{R}^n, g : \mathbb{R}^n \rightarrow \mathbb{R}^k$, then $D(g \circ f)(x) = Dg(f(x)) \cdot Df(x)$.
- **Attention!** If $f(\theta) : \mathbb{R}^d \rightarrow \mathbb{R}$, then $D_\theta f(\theta)$ is a row vector ($D_\theta f(\theta) = \nabla f(\theta)^T$).

6.9 Neural Tangent Kernel: definition

- We want to study the dynamics of a neural network $F_\theta : \mathbb{R}^{N_0} \rightarrow \mathbb{R}$ during Gradient Descent.
- We denote the empirical loss on the training set $D = ((x_i, y_i) \in \mathbb{R}^{N_0} \times \mathbb{R}, 1 \leq i \leq m)$ by

$$\mathcal{L}(\theta, D) = \frac{1}{m} \sum_{i=1}^m \ell(F_\theta(x_i), y_i),$$

where $\ell(\hat{y}, y)$ is a given loss function.

- For infinitesimally small gradient steps, gradient descent becomes a **gradient flow**:

$$D_t \theta_t = -D_\theta \mathcal{L}(\theta_t, D)^T,$$

6.9 Neural Tangent Kernel: definition

- The empirical training loss evolves according to:

$$D_t \mathcal{L}(\theta_t, D) = D_\theta \mathcal{L}(\theta_t, D) \cdot D_t \theta_t \quad (\text{chain rule})$$

$$= -D_\theta \mathcal{L}(\theta_t, D) \cdot D_\theta \mathcal{L}(\theta_t, D)^T \quad (\text{gradient flow definition})$$

Using the chain rule, we have that $D_\theta \mathcal{L}(\theta_t, D) = \frac{1}{m} \sum_{i=1}^m D_{\hat{y}} \ell(F_{\theta_t}(x_i), y_i) \cdot D_\theta F_{\theta_t}(x_i)$ and substituting above we obtain:

$$\begin{aligned} D_t \mathcal{L}(\theta_t, D) &= -\frac{1}{m^2} \sum_{i=1}^m \sum_{j=1}^m D_{\hat{y}} \ell(F_{\theta_t}(x_i), y_i) \cdot \underbrace{D_\theta F_{\theta_t}(x_i) \cdot (D_\theta F_{\theta_t}(x_j))^T}_{K(x_i, x_j; \theta_t)} (D_{\hat{y}} \ell(F_{\theta_t}(x_j), y_j))^T \\ &=: -\|D_{\hat{y}} \ell(F_{\theta_t}(\cdot), \cdot)\|_{K(\cdot, \cdot; \theta_t), D}^2 \end{aligned}$$

where $K(\cdot, \cdot; \theta) : \mathbb{R}^{N_0} \times \mathbb{R}^{N_0} \rightarrow \mathbb{R}$ is a **kernel** called the **Neural Tangent Kernel (NTK)** and $\|\cdot\|_{K, D}$ is the kernel norm in the RKHS associated to K .

6.9 Neural Tangent Kernel: definition

A few remarks...

- Assume $K(\cdot, \cdot; \theta_t)$ is constant (i.e. does not depend on θ_t) and positive-definite when evaluated on the dataset D , then:
 - the loss $\mathcal{L}$ decreases in time under the gradient flow (because $D_t \mathcal{L}(\theta_t, D) \leq 0$),
 - if $\hat{y} \mapsto \ell(\hat{y}, y_i)$ are convex and bounded from below, then $\mathcal{L}(\theta_t, D)$ converges to its global minimum.
- Unfortunately, there are two problems:
 - $K(\cdot, \cdot; \theta_t)$ is random (the parameters θ_t are initialized randomly),
 - $K(\cdot, \cdot; \theta_t)$ depends on θ_t (because $\theta \mapsto F_\theta(\cdot)$ is non-linear) and θ_t changes in time according to the gradient flow (hence the term **tangent** kernel).
- Luckily it can be shown [Jacot et al., 2018] that in the **infinite width limit** (i.e. as $N_1, \dots, N_{L-1} \rightarrow \infty$) the kernel $K(\cdot, \cdot; \theta_0)$ converges in probability to a deterministic kernel, which depends only on ρ and L , and therefore stays constant during training.
- This result also holds for neural networks with multivariate outputs, i.e. $F_\theta : \mathbb{R}^{N_0} \rightarrow \mathbb{R}^{N_L}$. In this case, though, the Neural Tangent Kernel is a **matrix-valued kernel** $K(\cdot, \cdot; \theta) : \mathbb{R}^{N_0} \times \mathbb{R}^{N_0} \rightarrow \mathbb{R}^{N_L} \times \mathbb{R}^{N_L}$ (see Problem Set 5, Ex. 3, 4, and 5).

6.9 Neural Tangent Kernel: definition

Some preliminary definitions...

Definition 1 (Gaussian Process (GP))

A family of random variables $\{f(x)\}_{x \in \mathcal{X}}$ is a **Gaussian Process** (GP) if for all $m \in \mathbb{N}$ and all $x_1, \dots, x_m \in \mathcal{X}$, $(f(x_1), \dots, f(x_m))$ is a multivariate Gaussian random vector.

Every GP is fully characterized by:

- the **mean function** $\mu : \mathcal{X} \rightarrow \mathbb{R}$, such that $\mu(x) = \mathbb{E}[f(x)]$ for all $x \in \mathcal{X}$, and
 - the **covariance kernel** $C : \mathcal{X} \times \mathcal{X} \rightarrow \mathbb{R}$, such that $\text{Cov}(f(x), f(x')) = C(x, x')$, for all $x, x' \in \mathcal{X}$.
- We write $f \sim \mathcal{GP}(\mu, C)$ for a Gaussian Process with mean μ and covariance kernel C .

6.9 Neural Tangent Kernel: convergence at initialization

- We focus on the case of real-valued shallow neural networks, i.e. neural networks of the form $F_\theta : \mathbb{R}^{N_0} \rightarrow \mathbb{R}$ with architecture $(N_0, N_1, 1)$.
- The general proof for a deep network with multivariate output $F_\theta : \mathbb{R}^{N_0} \rightarrow \mathbb{R}^{N_L}$ uses the same ideas and is presented in Problem Set 5.
- We are interested in the *infinite width limit* $N_1 \rightarrow \infty$, so it is convenient to rescale the network parameters:

$$\begin{cases} \tilde{a}_i^{(k)} &= \sum_{j=1}^{N_{k-1}} \frac{1}{\sqrt{N_{k-1}}} W_{ij}^{(k)} a_j^{(k-1)} + \beta b_i^{(k)} \\ a_i^{(k)} &= \rho(\tilde{a}_i^{(k)}) \end{cases}$$

for $k = 1, 2$ and for some $\beta > 0$.

- We will assume throughout that ρ is Lipschitz, twice-differentiable, with bounded second derivative.

6.9 Neural Tangent Kernel: convergence at initialization

Theorem 2 ([Jacot et al., 2018, Proposition 1])

Let $F_{\theta_0} : \mathbb{R}^{N_0} \rightarrow \mathbb{R}$ be a shallow neural network with parameter initialization $\theta_0 \sim \mathcal{N}(0, I_{P \times P})$, where P is the number of parameters.

Then as $N_1 \rightarrow \infty$ the function F_{θ_0} converges to a Gaussian process with covariance kernel C given by:

$$C(x, x') = \mathbb{E}_{f \sim \mathcal{GP}(0, \Sigma)} [\rho(f(x))\rho(f(x'))] + \beta^2, \quad \text{for all } x, x' \in \mathbb{R}^{N_0},$$

where

$$\Sigma(x, x') = \frac{x^T \cdot x'}{N_0} + \beta^2, \quad \text{for all } x, x' \in \mathbb{R}^{N_0}.$$

6.9 Neural Tangent Kernel: convergence at initialization

Proof.

- At initialization, the neural network output is given by

$$F_{\theta_0}(x) = \sum_{j=1}^{N_1} \frac{1}{\sqrt{N_1}} W_{1,j}^{(2)} \rho(\tilde{a}_j^{(1)}(x)) + \beta b^{(2)},$$

where $W^{(2)} \in \mathbb{R}^{1 \times N_1}$ and $b^{(2)} \in \mathbb{R}$ are the output layer parameters and $\tilde{a}^{(1)}$ are the pre-activations of the hidden layer, given by:

$$\tilde{a}_k^{(1)}(x) = \sum_{j=1}^{N_0} \frac{1}{\sqrt{N_0}} W_{k,j}^{(1)} x_j + \beta b_k^{(1)}, \quad k = 1, \dots, N_1.$$

6.9 Neural Tangent Kernel: convergence at initialization

Proof.

Due to the Gaussian parameter initialization, each pre-activation coordinate function $\{\tilde{a}_k^{(1)}(x)\}_{x \in \mathbb{R}^{N_0}}$ is a Gaussian process with covariance kernel:

$$\begin{aligned}\mathbb{E} \left[\tilde{a}_k^{(1)}(x) \tilde{a}_k^{(1)}(x') \right] &= \mathbb{E} \left[\left(\sum_{i=1}^{N_0} \frac{1}{\sqrt{N_0}} W_{k,i}^{(1)} x_i + \beta b_k^{(1)} \right) \left(\sum_{l=1}^{N_0} \frac{1}{\sqrt{N_0}} W_{k,l}^{(1)} x'_l + \beta b_k^{(1)} \right) \right] \\&= \sum_{i=1}^{N_0} \sum_{l=1}^{N_0} \frac{1}{N_0} \mathbb{E} \left[W_{k,i}^{(1)} W_{k,l}^{(1)} \right] x_i x'_l + \beta^2 \mathbb{E} \left[b_k^{(1)} b_k^{(1)} \right] \\&= \sum_{i=1}^{N_0} \sum_{l=1}^{N_0} \frac{1}{N_0} \delta_{i,l} x_i x'_l + \beta^2 \mathbb{E} \left[b_k^{(1)} b_k^{(1)} \right] \\&= \frac{x^T \cdot x'}{N_0} + \beta^2 =: \Sigma(x, x')\end{aligned}$$

6.9 Neural Tangent Kernel: convergence at initialization

Proof.

Analogously we have that, conditionally on the preactivations $\tilde{a}^{(1)}$, also the network output $F_{\theta_0}(x)$ is a Gaussian process with covariance kernel given by:

$$\mathbb{E} \left[F_{\theta_0}(x) F_{\theta_0}(x') \mid \tilde{a}^{(1)} \right] = \frac{1}{N_1} \sum_{k=1}^{N_1} \rho(\tilde{a}_k^{(1)}(x)) \rho(\tilde{a}_k^{(1)}(x')) + \beta^2.$$

Since the Gaussian processes $(\tilde{a}_k^{(1)}(x), k = 1, \dots, N_1)$ are independent and identically distributed, the terms $\rho(\tilde{a}_k^{(1)}(x)) \rho(\tilde{a}_k^{(1)}(x'))$ are iid random variables and by the law of large numbers their normalized sum converges as $N_1 \rightarrow \infty$ to their common expectation:

$$\frac{1}{N_1} \sum_{k=1}^{N_1} \rho(\tilde{a}_k^{(1)}(x)) \rho(\tilde{a}_k^{(1)}(x')) \rightarrow \mathbb{E}_{f \sim \mathcal{GP}(0, \Sigma)} [\rho(f(x)) \rho(f(x'))],$$

which yields exactly the expression given in the statement of the theorem. □

6.9 Neural Tangent Kernel: convergence at initialization

Theorem 3 (Convergence at initialization [Jacot et al., 2018, Theorem 1])

Let $F_{\theta_0} : \mathbb{R}^{N_0} \rightarrow \mathbb{R}$ be a shallow neural network with parameter initialization $\theta_0 \sim \mathcal{N}(0, I_{P \times P})$, where P is the number of parameters, and let $K(x, x'; \theta_0) := D_{\theta} F_{\theta_0}(x) \cdot D_{\theta} F_{\theta_0}(x')^T$ denote the Neural Tangent Kernel (NTK) of the network at initialization.

Then we have that:

$$K(x, x'; \theta_0) \xrightarrow{\mathbb{P}} \tilde{K}(x, x') \text{ as } N_1 \rightarrow \infty,$$

where $\tilde{K} : \mathbb{R}^{N_0} \times \mathbb{R}^{N_0} \rightarrow \mathbb{R}$ is a deterministic kernel defined as:

$$\tilde{K}(x, x') := \Sigma(x, x') \dot{\Sigma}(x, x') + C(x, x'),$$

with $\Sigma(x, x')$ and $C(x, x')$ as in Theorem 2 and $\dot{\Sigma}(x, x') := \mathbb{E}_{f \sim \mathcal{GP}(0, \Sigma)} [\rho'(f(x)) \rho'(f(x'))]$.

- The limiting kernel $\tilde{K}$ depends only on ρ .
- $\implies$ It does *not* depend on the (random) parameters θ and remains constant during training.

6.9 Neural Tangent Kernel: convergence at initialization

Proof.

- By definition, the NTK is given by:

$$\begin{aligned} K(x, x'; \theta) &= D_{\theta} F_{\theta}(x) \cdot D_{\theta} F_{\theta}(x')^T \\ &= \sum_{\theta_p} D_{\theta_p} F_{\theta}(x) D_{\theta_p} F_{\theta}(x'), \end{aligned} \quad (\text{☕})$$

where we sum over all univariate parameters $\theta_p \in (W^{(1)}, b^{(1)}, W^{(2)}, b^{(2)})$ of the network.

- We can split the sum in (☕) over univariate parameters of the first layer $\tilde{\theta}_p \in (W^{(1)}, b^{(1)})$ and of the second layer $\theta'_p \in (W^{(2)}, b^{(2)})$.

6.9 Neural Tangent Kernel: convergence at initialization

Proof.

- We first focus on the parameters of the first layer and compute $\sum_{\tilde{\theta}_p} D_{\tilde{\theta}_p} F_{\theta}(x) D_{\tilde{\theta}_p} F_{\theta}(x')$ using the chain rule with respect to the pre-activations $\tilde{a}^{(1)}$, obtaining:

$$\begin{aligned} \sum_{\tilde{\theta}_p} D_{\tilde{\theta}_p} F_{\theta}(x) D_{\tilde{\theta}_p} F_{\theta}(x') &= \sum_{\tilde{\theta}_p} D_{\tilde{a}^{(1)}} F_{\theta}(x) D_{\tilde{\theta}_p} \tilde{a}^{(1)}(x) D_{\tilde{a}^{(1)}} F_{\theta}(x') D_{\tilde{\theta}_p} \tilde{a}^{(1)}(x') \\ &= \sum_{\tilde{\theta}_p} \left(\sum_{k=1}^{N_1} D_{\tilde{a}_k^{(1)}} F_{\theta}(x) D_{\tilde{\theta}_p} \tilde{a}_k^{(1)}(x) \right) \left(\sum_{j=1}^{N_1} D_{\tilde{a}_j^{(1)}} F_{\theta}(x') D_{\tilde{\theta}_p} \tilde{a}_j^{(1)}(x') \right) \\ &= \sum_{k=1}^{N_1} \sum_{j=1}^{N_1} \sum_{\tilde{\theta}_p} D_{\tilde{a}_k^{(1)}} F_{\theta}(x) D_{\tilde{\theta}_p} \tilde{a}_k^{(1)}(x) D_{\tilde{a}_j^{(1)}} F_{\theta}(x') D_{\tilde{\theta}_p} \tilde{a}_j^{(1)}(x') \end{aligned} \tag{1}$$

6.9 Neural Tangent Kernel: convergence at initialization

Proof.

Computing the term in blue yields:

$$\begin{aligned}\sum_{\tilde{\theta}_p} D_{\tilde{\theta}_p} \tilde{a}_k^{(1)}(x) D_{\tilde{\theta}_p} \tilde{a}_j^{(1)}(x') &= \sum_{i=1}^{N_1} \sum_{l=1}^{N_0} \frac{\partial \tilde{a}_k^{(1)}(x)}{\partial W_{i,l}^{(1)}} \frac{\partial \tilde{a}_j^{(1)}(x')}{\partial W_{i,l}^{(1)}} + \sum_{i=1}^{N_1} \frac{\partial \tilde{a}_k^{(1)}(x)}{\partial b_i^{(1)}} \frac{\partial \tilde{a}_j^{(1)}(x')}{\partial b_i^{(1)}} \\ &= \frac{1}{N_0} \sum_{i=1}^{N_1} \sum_{l=1}^{N_0} \delta_{k,i} x_l \delta_{j,i} x'_l + \sum_{i=1}^{N_1} \delta_{i,k} \beta \delta_{i,j} \beta \\ &= \left(\frac{x^T \cdot x'}{N_0} + \beta^2 \right) \delta_{k,j} = \Sigma(x, x') \delta_{k,j}\end{aligned}$$

6.9 Neural Tangent Kernel: convergence at initialization

Proof.

Substituting the blue term in Equation (1), we then obtain:

$$\begin{aligned}\sum_{\tilde{\theta}_p} D_{\tilde{\theta}_p} F_{\theta}(x) D_{\tilde{\theta}_p} F_{\theta}(x') &= \sum_{k=1}^{N_1} D_{\tilde{a}_k^{(1)}} F_{\theta}(x) D_{\tilde{a}_k^{(1)}} F_{\theta}(x') \Sigma(x, x') \\ &= \frac{1}{N_1} \sum_{k=1}^{N_1} W_{1,k}^{(2)} \rho'(\tilde{a}_k^{(1)}(x)) W_{1,k}^{(2)} \rho'(\tilde{a}_k^{(1)}(x')) \Sigma(x, x')\end{aligned}$$

The pre-activations $(\tilde{a}_k^{(1)}(x), k = 1, \dots, N_1)$ are iid Gaussian processes $\mathcal{GP}(0, \Sigma)$ (see proof of Theorem 2) independent of the second layer parameters, therefore the law of large numbers yields:

$$\begin{aligned}\frac{1}{N_1} \sum_{k=1}^{N_1} W_{1,k}^{(2)} \rho'(\tilde{a}_k^{(1)}(x)) W_{1,k}^{(2)} \rho'(\tilde{a}_k^{(1)}(x')) \Sigma(x, x') &\rightarrow \mathbb{E}[W_{1,k}^{(2)} W_{1,k}^{(2)}] \mathbb{E} \left[\rho'(\tilde{a}_k^{(1)}(x)) \rho'(\tilde{a}_k^{(1)}(x')) \right] \Sigma(x, x') \\ &= \mathbb{E}_{f \sim \mathcal{GP}(0, \Sigma)} [\rho'(f(x)) \rho'(f(x'))] \Sigma(x, x') \\ &=: \dot{\Sigma}(x, x') \Sigma(x, x')\end{aligned}$$

6.9 Neural Tangent Kernel: convergence at initialization

Proof.

- Finally, we focus on the parameters of the second layer and compute $\sum_{\theta'_p} D_{\theta'_p} F_{\theta}(x) D_{\theta'_p} F_{\theta}(x')$.
- In this case, there's no need to apply the chain rule, and we obtain directly:

$$\begin{aligned} \sum_{\theta'_p} D_{\theta'_p} F_{\theta}(x) D_{\theta'_p} F_{\theta}(x') &= \frac{1}{N} \sum_{l=1}^{N_1} D_{W_{1,l}^{(2)}} F_{\theta}(x) D_{W_{1,l}^{(2)}} F_{\theta}(x') + D_{b_1^{(2)}} F_{\theta}(x) D_{b_1^{(2)}} F_{\theta}(x') \\ &= \frac{1}{N} \sum_{l=1}^{N_1} \rho(\tilde{a}_l^{(1)}(x)) \rho(\tilde{a}_l^{(1)}(x')) + \beta^2 \end{aligned}$$

A last application of the law of large numbers yields:

$$\frac{1}{N} \sum_{l=1}^{N_1} \rho(\tilde{a}_l^{(1)}(x)) \rho(\tilde{a}_l^{(1)}(x')) + \beta^2 \rightarrow \mathbb{E}_{f \sim \mathcal{GP}(0, \Sigma)} [\rho(f(x)) \rho'(f(x))] + \beta^2 =: C(x, x').$$

The claim follows by summing together the terms of the first and second layer parameters.



6.9 Neural Tangent Kernel: convergence during training

We mention (without proof) that the following stronger result holds:

Theorem 4 (Convergence during training [Jacot et al., 2018, Theorem 2])

Let $F_\theta : \mathbb{R}^{N_0} \rightarrow \mathbb{R}$ be a neural network with $L + 1$ layers initialized with $\theta_0 \sim \mathcal{N}(0, I_{P \times P})$, then for any $T > 0$ such that $\int_0^T \frac{1}{m} \sum_{i=1}^m \|D_{\hat{y}} \ell(F_{\theta_t}(x_i), y_i)\|_2^2 dt$ stays stochastically bounded, we have that

$$\sup_{t \in [0, T]} \|K(x, x'; \theta_t) - \tilde{K}(x, x')\| \xrightarrow{\mathbb{P}} 0 \text{ as } N_1, \dots, N_{L-1} \rightarrow \infty$$

where K is the NTK and $\tilde{K} : \mathbb{R}^{N_0} \times \mathbb{R}^{N_0} \rightarrow \mathbb{R}$ is the limiting kernel as in Theorem 3.

- The NTK of a large network remains “asymptotically constant” during training
- The proof relies on the following facts:

$$\|W_i^{(L)}(t) - W_i^{(L)}(0)\|_2 = O\left(1/\sqrt{N_{L-1}}\right), \text{ where } W_i^{(L)} \text{ is the } i\text{-th column of } W,$$

$$\frac{1}{m} \sum_{i=1}^m \|F_{\tilde{\theta}_{t,k}}^{(L)}(x_i) - F_{\tilde{\theta}_{0,k}}^{(L)}(x_i)\|_2^2 = O\left(1/\sqrt{N_{L-1}}\right)$$

$\implies$ the weights of overparametrized networks move very little during training.
This phenomenon is known as **lazy training** and it has been observed empirically.

6.10 Convergence to the minimum-norm interpolator

- We have seen that large neural networks behave like kernel models. But what kind of kernel models?
- Consider a univariate regression problem with loss $\ell(\hat{y}, y) = (\hat{y} - y)^2$.
- Then in the infinite width limit the neural network $F_\theta(\cdot)$ evolves under gradient flow as:

$$\begin{aligned} D_t F_{\theta_t}(\cdot) &= D_\theta F_{\theta_t}(\cdot) \cdot D_t \theta_t \\ &= -D_\theta F_{\theta_t}(\cdot) \cdot D_\theta \mathcal{L}(\theta_t, D)^T && \text{(gradient flow } D_t \theta_t = -D_\theta \mathcal{L}(\theta_t, D)^T) \\ &= -\frac{1}{m} \sum_{i=1}^m \underbrace{D_\theta F_{\theta_t}(\cdot) D_\theta F_{\theta_t}(x_i)^T}_{\tilde{K}^{(L)}(\cdot, x_i)} D_{\hat{y}} \ell(F_{\theta_t}(x_i), y_i)^T \\ &&& (K^{(L)}(\cdot, x_i; \theta_t) = \tilde{K}^{(L)}(\cdot, x_i) \text{ by Theorem 4)} \\ &= -\frac{2}{m} \sum_{i=1}^m \underbrace{D_\theta F_{\theta_t}(\cdot) D_\theta F_{\theta_t}(x_i)^T}_{\tilde{K}^{(L)}(\cdot, x_i)} (F_{\theta_t}(x_i) - y_i) && (\ell(\hat{y}, y) = (\hat{y} - y)^2) \end{aligned}$$

6.10 Convergence to the minimum-norm interpolator

- One can check that this differential equation has the following solution (see Problem Set 5, Ex 6):

$$F_{\theta_t}(\cdot) = F_{\theta_0}(\cdot) + \sum_{i=1}^m \sum_{j=1}^m \sum_{k=1}^m \tilde{K}^{(L)}(\cdot, x_i) \bar{K}_{i,j}^{-1} (\delta_{j,k} - (e^{-2\bar{K}t/m})_{j,k}) (y_k - F_{\theta_0}(x_k)),$$

where $\bar{K}_{i,j} := \tilde{K}^{(L)}(x_i, x_j)$ (which is invertible if $\tilde{K}^{(L)}(\cdot, \cdot)$ is positive definite on the dataset) and $e^{-2\bar{K}t/m}$ is the matrix exponential of the matrix $-2\bar{K}t/m$.

- For $t \rightarrow \infty$, the solution is:

$$F_{\theta_\infty}(\cdot) = F_{\theta_0}(\cdot) + \sum_{i=1}^m \sum_{j=1}^m \tilde{K}^{(L)}(\cdot, x_i) \bar{K}_{i,j}^{-1} (y_j - F_{\theta_0}(x_j))$$

6.10 Convergence to the minimum-norm interpolator

By rearranging the terms:

$$F_{\theta_\infty}(\cdot) = \underbrace{\sum_{i=1}^m \sum_{j=1}^m \tilde{K}^{(L)}(\cdot, x_i) \bar{K}_{i,j}^{-1} y_j}_{\text{deterministic term (A)}} + \underbrace{F_{\theta_0}(\cdot) - \sum_{i=1}^m \sum_{j=1}^m \tilde{K}^{(L)}(\cdot, x_i) \bar{K}_{i,j}^{-1} F_{\theta_0}(x_j)}_{\text{random term (B)}}$$

- (A) is the solution to the minimum-norm interpolation problem in the RKHS $\mathcal{H}$ with kernel $\tilde{K}^{(L)}$ (see Problem Set 5, Ex 7), i.e.:

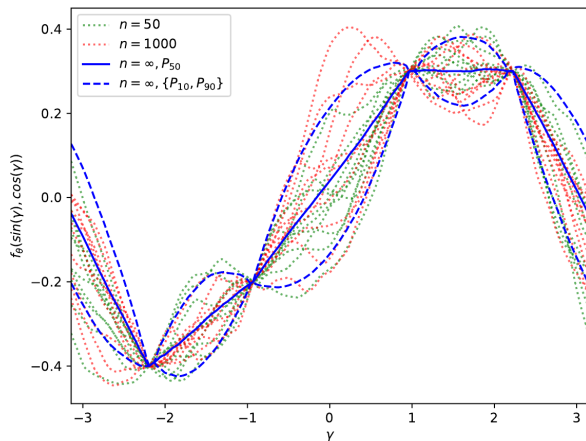
$$(A) = \arg \min_{\substack{f \in \mathcal{H} \\ f(x_i) = y_i}} \|f\|_{\mathcal{H}}$$

Notice that (A) is also the solution of Ridge regression with $\lambda \downarrow 0$.

- (B) is a centered Gaussian process that vanishes on each datapoint x_k :

$$F_{\theta_0}(x_k) - \sum_{i=1}^m \sum_{j=1}^m \tilde{K}^{(L)}(x_k, x_i) \bar{K}_{i,j}^{-1} F_{\theta_0}(x_j) = F_{\theta_0}(x_k) - \sum_{j=1}^m \delta_{k,j} F_{\theta_0}(x_j) = 0$$

6.10 Convergence to the minimum-norm interpolator



- **Task^a**: approximate $F^* : \mathbb{R}^2 \rightarrow \mathbb{R}$, $F^*(x) = x_1 x_2$ on $x = (\cos(\gamma), \sin(\gamma))$ with squared loss on a training set of four points.
- We train networks of depth $L = 4$ and two different widths $N = 50$ and $N = 1000$ for 1000 GD steps and for ten different parameter initializations.
- Each network is trained and compared to the median, lower and upper deciles (P_{50}, P_{10} and P_{90} , resp.) of the **limiting Gaussian process** ($N = \infty$) with kernel $\tilde{K}^{(4)}$.

^aImage taken from [Jacot et al., 2018]

6.11 Generalization in the overparametrized regime

- We have seen that infinite-width (=overparametrized) neural networks behave like kernel models, specifically like minimum-norm kernel interpolators in the NTK RKHS.
- This connection allows us to exploit many well-known results on the **generalization properties** of minimum-norm kernel interpolators [Belkin et al., 2018] [Liang and Rakhlin, 2020] [Liang et al., 2020] [Ghorbani et al., 2021] [Li, 2021]
- To understand these results in a friendly way, we focus on the simpler setting of linear regression (the L -th layer of the network) on random features (the output of the first $L - 1$ layers with random initialization) in the overparametrized regime.

6.11 Generalization in the overparametrized regime

- Consider the same setting as linear regression in Lecture 2.
- Let $X = (X_1, \dots, X_p)$ be $\mathbb{R}^p$ -valued random vector with covariance Σ . The true model is:

$$Y = X^T \beta + \varepsilon$$

with $\varepsilon \sim \mathcal{N}(0, \sigma^2)$ independent of X .

- We have a training set $D = (X'_{i,1}, \dots, X'_{i,p}, Y'_i), 1 \leq i \leq m)$ of m iid realizations of $(X_1, \dots, X_p, Y)$, with design matrix:

$$A = \begin{pmatrix} X'_{1,1} & \cdots & X'_{1,p} \\ X'_{2,1} & \cdots & X'_{2,p} \\ \vdots & \ddots & \vdots \\ X'_{m,1} & \cdots & X'_{m,p} \end{pmatrix}$$

- Let $\hat{\beta}$ be the least squares estimator with minimum $\|\cdot\|_2$ norm, i.e. $\hat{\beta} = (A^T A)^\dagger A^T Y'$.

6.11 Generalization in the overparametrized regime

- Denote by $\hat{f}_D(x) = x^T \hat{\beta}$ the estimated model and by $f^*(x) = x^T \beta$ the true model.
- We have the following bias-variance decomposition conditionally on the design matrix A (see Problem Set 5, Ex 8):

$$\mathbb{E} \left[(\hat{f}_D(X) - f^*(X))^2 \mid A \right] = \underbrace{\beta^T \Pi \Sigma \Pi \beta}_{B_A = \text{bias}^2} + \underbrace{\frac{\sigma^2}{m} \text{Tr}(\hat{\Sigma}^\dagger \Sigma)}_{V_A = \text{variance}}$$

where $\hat{\Sigma} = \frac{1}{m} A^T A$ and $\Pi = I_p - \hat{\Sigma}^\dagger \hat{\Sigma}$.

Theorem 5 ([Hastie et al., 2022])

Assume that $(X_1, \dots, X_p)$ has iid compenents with unit variance and finite fifth-moments. Then as $m, p \rightarrow \infty$ with $p/m \rightarrow \gamma \in (0, 1) \cup (1, +\infty)$ the variance of the minimum-norm least squares estimator converges almost surely to:

$$\lim_{\substack{m, p \rightarrow \infty \\ p/m \rightarrow \gamma}} V_A = \sigma^2 \frac{1 - \max\{1 - \gamma, 0\}}{|1 - \gamma|}$$

Random matrix theory: the Marchenko-Pastur law

We will need a fundamental result of random matrix theory, which we present here without proof.

Theorem 6 (Marchenko-Pastur law)

Let B be a $p \times m$ random matrix with iid entries with zero mean and variance σ^2 .

Let $Y_m = \frac{1}{m}BB^T$ and consider its **empirical spectral measure** $\mu_{Y_m} = \frac{1}{p} \sum_{i=1}^p \delta_{\lambda_i}$, where $\lambda_1, \lambda_2, \dots, \lambda_p$ are the (random) eigenvalues of Y_m .

Then as $p, m \rightarrow \infty$ with $p/m \rightarrow \gamma \in (0, 1)$ the random measure μ_{Y_m} converges weakly to a deterministic measure μ_γ :

$$\mu_{Y_m}(ds) \longrightarrow \mu_\gamma(ds) := \frac{1}{2\pi\sigma^2} \frac{\sqrt{(\gamma_+ - s)(s - \gamma_-)}}{\gamma s} \mathbf{1}_{[\gamma_-, \gamma_+]}(s) ds$$

where $\gamma_\pm = \sigma^2(1 \pm \sqrt{\gamma})^2$.

More precisely^a, for every bounded, continuous function f :

$$\lim_{\substack{p, m \rightarrow \infty \\ p/m \rightarrow \gamma}} \frac{1}{p} \sum_{i=1}^p f(\lambda_i) = \int f(s) \mu_\gamma(ds), \quad \mathbb{P} - a.s.$$

^aMarchenko and Pastur proved convergence in probability [Marchenko and Pastur, 1967]. The result was strengthened to almost sure convergence in [Silverstein, 1995]

6.11 Generalization in the overparametrized regime

Proof of Theorem 5.

We first deal with the case $\gamma \in (0, 1)$.

- Since $\Sigma = I_p$, the expression for the variance is $V_A = \frac{\sigma^2}{m} \text{Tr}(\hat{\Sigma}^\dagger) = \frac{\sigma^2}{m} \text{Tr} \left(\left(\frac{A^T A}{m} \right)^\dagger \right)$.
- The non-zero eigenvalues of $(A^T A)^\dagger / m$ are just the reciprocals of the non-zero eigenvalues of $(A^T A)/m$, therefore:

$$V_A = \frac{\sigma^2}{m} \text{Tr} \left(\left(\frac{A^T A}{m} \right)^\dagger \right) = \frac{\sigma^2}{m} p \int_0^\infty \frac{1}{s} \mu_{A^T A/m}(ds) \xrightarrow{a.s.} \sigma^2 \gamma \int_0^\infty \frac{1}{s} \mu_\gamma(ds),$$

where μ_γ is the Marchenko-Pastur law.

6.11 Generalization in the overparametrized regime

Proof.

- The integral is exactly the Stieltjes transform of μ_γ , $m(z) = \int_0^\infty \frac{1}{s-z} \mu_\gamma(ds)$, evaluated at $z = 0$. An expression for $m(z)$ exists for $\text{Re}(z) > 0$:

$$m(-z) = \frac{-(1 - \gamma + z) + \sqrt{(1 - \gamma + z)^2 + 4\gamma z}}{2\gamma z}.$$

By taking the limit for $z \rightarrow 0^+$ (use L'Hôpital's rule) we obtain $V_A \xrightarrow{a.s.} \sigma^2 \gamma \frac{1}{1-\gamma}$.

6.11 Generalization in the overparametrized regime

Proof.

We now deal with the case $\gamma \in (1, \infty)$.

- We start by noticing that $A^T A$ and AA^T have the same eigenvalues and therefore $\lambda_i(A^T A/m) = \frac{p}{m} \lambda_i(AA^T/p)$:

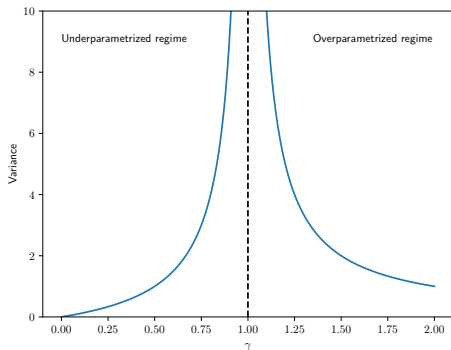
$$\begin{aligned} V_A &= \frac{\sigma^2}{m} \text{Tr} \left(\left(\frac{A^T A}{m} \right)^\dagger \right) = \frac{\sigma^2}{m} \sum_{i|\lambda_i > 0} \frac{1}{\lambda_i(A^T A/m)} \\ &= \frac{\sigma^2}{p} \sum_{i|\lambda_i > 0} \frac{1}{\lambda_i(AA^T/p)} = \sigma^2 \frac{m}{p} \int_0^\infty \frac{1}{s} \mu_{AA^T/p}(ds) \end{aligned}$$

- As $p/m \rightarrow \gamma$ we have that $m/p \rightarrow 1/\gamma \in (0, 1)$ and $\mu_{AA^T/m}$ converges to the Marchenko-Pastur law $\mu_{1/\gamma}$. Using the same computation from the previous step, we obtain:

$$V_A \xrightarrow{a.s.} \sigma^2 \frac{1}{\gamma} \frac{1}{1 - \frac{1}{\gamma}} = \frac{\sigma^2}{\gamma - 1}.$$

6.11 Generalization in the overparametrized regime

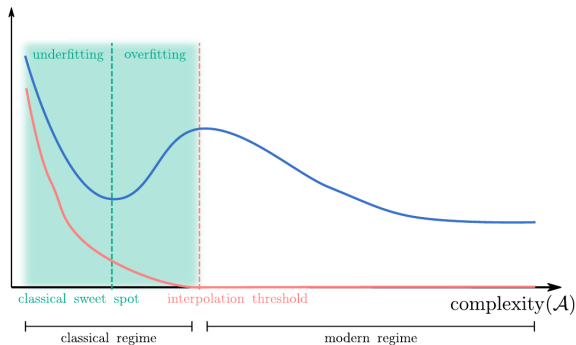
A few remarks about the result...



- γ quantifies model complexity
 $\gamma \gg 1 \Rightarrow p \gg m \Rightarrow$ complex model
- Variance increases in model complexity for $\gamma \in (0, 1)$ (classical bias-variance tradeoff)
- Variance decreases in model complexity for $\gamma > 1$
- In the overparametrized regime the minimum-norm least squares solution uses the additional parameters to minimize the norm $\Rightarrow$ maximize the margin $\Rightarrow$ generalize well
- [Hastie et al., 2022] prove that for misspecified models (i.e. true model is not linear) the generalization error can have a **global** minimum in the **overparametrized** regime

Conclusions

Practical consequences for training neural networks:



- Variance decreases as soon as model complexity passes the interpolation threshold
- The **test error** can decrease even after the **train error** has reached zero
- The best generalization error may be in the overparametrized regime (i.e. not the classical "bias-variance" sweet spot)
- This behavior is referred to as the *double descent curve* or *benign overfitting*.

References I

-  Arora, S., Ge, R., Neyshabur, B., and Zhang, Y. (2018).
Stronger generalization bounds for deep nets via a compression approach.
In International Conference on Machine Learning, pages 254–263. PMLR.
-  Bartlett, P. (1998).
The sample complexity of pattern classification with neural networks: the size of the weights
is more important than the size of the network.
IEEE Transactions on Information Theory, 44(2):525–536.
-  Bartlett, P. L., Bousquet, O., and Mendelson, S. (2005).
Local rademacher complexities.
-  Belkin, M., Hsu, D., Ma, S., and Mandal, S. (2019).
Reconciling modern machine-learning practice and the classical bias–variance trade-off.
Proceedings of the National Academy of Sciences, 116(32):15849–15854.
-  Belkin, M., Ma, S., and Mandal, S. (2018).
To understand deep learning we need to understand kernel learning.
In International Conference on Machine Learning, pages 541–549. PMLR.

References II

-  Bousquet, O. and Elisseeff, A. (2002).
Stability and generalization.
The Journal of Machine Learning Research, 2:499–526.
-  Dziugaite, G. K. and Roy, D. M. (2017).
Computing nonvacuous generalization bounds for deep (stochastic) neural networks with many more parameters than training data.
arXiv preprint arXiv:1703.11008.
-  Ghorbani, B., Mei, S., Misiakiewicz, T., and Montanari, A. (2021).
Linearized two-layers neural networks in high dimension.
-  Hastie, T., Montanari, A., Rosset, S., and Tibshirani, R. J. (2022).
Surprises in high-dimensional ridgeless least squares interpolation.
Annals of statistics, 50(2):949.
-  Jacot, A., Gabriel, F., and Hongler, C. (2018).
Neural tangent kernel: Convergence and generalization in neural networks.
Advances in neural information processing systems, 31.

References III

-  Li, W. (2021).
Generalization error of minimum weighted norm and kernel interpolation.
SIAM Journal on Mathematics of Data Science, 3(1):414–438.
-  Liang, T. and Rakhlin, A. (2020).
Just interpolate: Kernel “ridgeless” regression can generalize.
-  Liang, T., Rakhlin, A., and Zhai, X. (2020).
On the multiple descent of minimum-norm interpolants and restricted lower isometry of kernels.
In Conference on Learning Theory, pages 2683–2711. PMLR.
-  Marchenko, V. A. and Pastur, L. A. (1967).
Distribution of eigenvalues for some sets of random matrices.
Matematicheskii Sbornik, 114(4):507–536.
-  Neyshabur, B., Bhojanapalli, S., McAllester, D., and Srebro, N. (2017).
Exploring generalization in deep learning.
Advances in neural information processing systems, 30.

References IV

-  Neyshabur, B., Tomioka, R., and Srebro, N. (2014).
In search of the real inductive bias: On the role of implicit regularization in deep learning.
arXiv preprint arXiv:1412.6614.
-  Poggio, T., Rifkin, R., Mukherjee, S., and Niyogi, P. (2004).
General conditions for predictivity in learning theory.
Nature, 428(6981):419–422.
-  Silverstein, J. W. (1995).
Strong convergence of the empirical distribution of eigenvalues of large dimensional random matrices.
Journal of Multivariate Analysis, 55(2):331–339.
-  Xu, H. and Mannor, S. (2012).
Robustness and generalization.
Machine learning, 86:391–423.
-  Zhang, C., Bengio, S., Hardt, M., Recht, B., and Vinyals, O. (2017).
Understanding deep learning requires rethinking generalization.